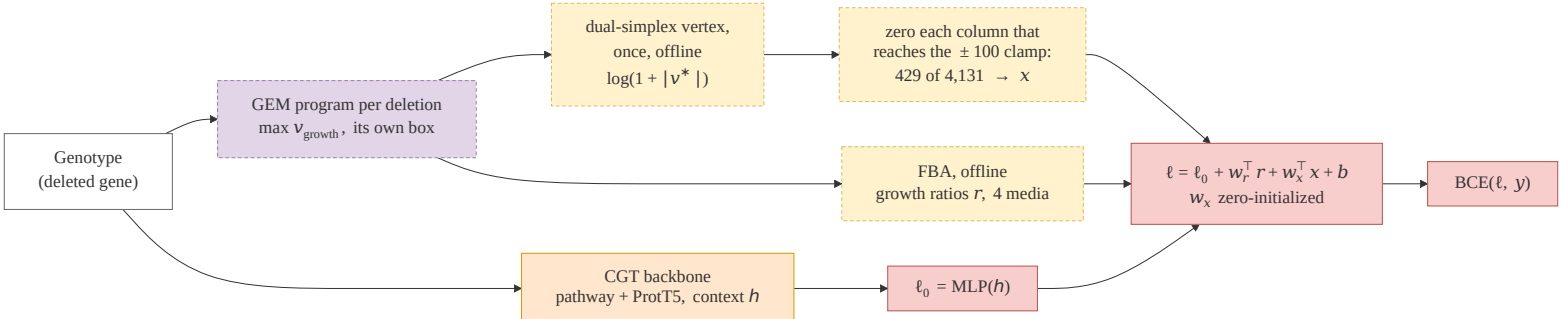

Genotype
(deleted gene)
GEM program per deletion
max v_{growth}, its own box
dual-simplex vertex,
once, offline
$\log(1 + |v^*|)$
zero each column that
reaches the ± 100 clamp:
429 of 4,131 → x
FBA, offline
growth ratios r, 4 media
CGT backbone
pathway + ProtT5, context h
ℓ_0 = MLP(h)
$\ell = \ell_0 + w_r^\top r + w_x^\top x + b$
w_x zero-initialized
BCE(ℓ, y)